SALMAN SHAWKAT KAZI

Software Engineer · Full-Stack Web Developer · AI & Machine Learning

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PROFESSIONAL SUMMARY

Motivated Software Engineer and Computer Science graduate (MSc, Nottingham Trent University; BSc with distinction, CGPA 9.49/10) with a strong foundation across the full software development lifecycle. Proficient in backend engineering (PHP/Laravel, Java), full-stack web development, machine learning, and cloud-deployed distributed systems (Docker, Microsoft Azure). Experienced in writing clean, maintainable, object-oriented code, designing RESTful APIs, working with SQL and NoSQL databases, and delivering CI/CD pipelines using Jenkins and GitHub. An active daily user of AI development tools Claude, ChatGPT, Google Gemini, GitHub Copilot, and Cursor to accelerate delivery, improve code quality, and stay at the forefront of modern engineering practices.

TECHNICAL SKILLS

Languages: PHP, Python, Java, JavaScript, TypeScript (familiar), C++, HTML5, CSS3

Backend & APIs: Laravel, Livewire, Eloquent ORM, RESTful API Design & Integration, Apache Tomcat, Microservices / SOA

Frontend & Web: React, Next.js, Tailwind CSS, Bootstrap, Alpine.js, Blade, Responsive Design

AI & Machine Learning: Keras, TensorFlow, Scikit-Learn, Pandas, NumPy, MLP, CNN, Gradient Boosted Trees, K-Means Clustering, Feature Engineering, LLM APIs (GPT-4, Gemini, Claude)

AI-Assisted Development: Claude (Anthropic), ChatGPT (OpenAI), Google Gemini, GitHub Copilot, Cursor daily use to accelerate development, generate boilerplate, debug complex issues, write tests, and improve code quality

Databases & SQL: MySQL, MongoDB Atlas (NoSQL), Firebase, SQLite, Relational Schema Design, SQL Query Writing & Optimisation

DevOps & Cloud: Docker (containerisation), Git / GitHub (version control), Jenkins (CI/CD pipelines), Microsoft Azure (Linux VM deployment), Environment Management

Architecture: MVC, SOLID Principles, Object-Oriented Design (OOP), Service-Oriented Architecture (SOA), Test-Driven Development (TDD), RBAC

Testing & QA: JUnit (automated pipelines), Apache JMeter (load testing), model evaluation (Silhouette Score, Adjusted Rand Index), TDD methodology

PROJECTS

Student Event Booking Platform | Distributed Service-Oriented System

Java 17 · Apache Tomcat · PHP · Laravel · Livewire · MongoDB Atlas · Docker · Microsoft Azure · JMeter

- Architected a production-grade, loosely coupled distributed system in Java 17, separating a RESTful API backend (Apache Tomcat) from a full-stack PHP/Laravel/Livewire presentation layer applying SOA and MVC design principles across a multi-service architecture.
- Integrated four third-party APIs (GeoNames, Google Maps, OpenWeather, Skiddle) to automate data enrichment, gaining hands-on experience of API design, authentication, and real-world integration workflows.
- **Debugged and resolved N+1 SQL query bottlenecks** using cursor pagination, lazy loading, and in-memory HashMap caching materially improving system throughput and resilience under load.
- **Stress-tested with Apache JMeter 1,000 concurrent users:** achieved 5-second response times and a 0.00% error rate across all GET and POST endpoints on a live Microsoft Azure deployment, validating production stability.
- Containerised the Java REST API backend with Docker and deployed to a live Azure Linux VM — demonstrating end-to-end ownership of build, containerisation, deployment, and operation.

CI/CD Pipeline & Automated Testing | Advanced Software Engineering Module

Jenkins · GitHub · Docker · JUnit · Automated Build, Test & Deployment Pipelines

- Designed and implemented CI/CD pipelines using Jenkins and GitHub, automating build, test, and deployment stages with quality gates enforced on every commit ensuring no code reached deployment without passing a full automated test suite.

- Integrated JUnit regression suites into the pipeline and managed Docker-based environment consistency across build and deployment stages, applying professional software engineering discipline to the full development lifecycle.

Accommodation Maintenance System | Native Android Application

Java · Android Studio · Firebase Auth · Firestore · Firebase Storage · CameraX · RBAC

- Developed a native Java Android application replacing paper-based maintenance workflows with a real-time digital system, implementing RBAC across three user roles (manager, tenant, staff) using Firebase Authentication.
- Built a custom CameraX module for photo capture, annotation, and attachment directly to maintenance requests improving issue accuracy and enabling remote pre-assessment by staff before site visits.
- Used Firestore real-time listeners to push live status updates to users demonstrating event-driven, real-time data architecture applicable to any high-frequency notification system.

Distributed Secure Cloud Storage System | Desktop Application

Java · JavaFX · Docker · JSch (SSH) · JUnit · Automated Testing Pipelines

- Architected a distributed, encrypted file storage system in Java across multiple isolated Docker containers addressing centralised single-point-of-failure risk with an automated encryption and distribution pipeline.
- Built a secure SSH remote administration terminal (JSch) and validated data integrity across all distributed nodes via automated JUnit pipelines, applying test-driven development to verify correctness in a distributed, failure-prone environment.

Business Workflow & Automation Platforms | Full-Stack Web Applications

PHP · Laravel · Livewire · MySQL · Tailwind CSS · Bootstrap · RBAC · Automated Notifications

- Delivered four full-stack business automation systems Inventory Management, Support Ticketing, Complaint Management, and Attendance Recording each converting real operational business requirements into clean, maintainable PHP/Laravel code with RBAC, audit logs, real-time Livewire interfaces, and automated notifications.
- Built automated CSV/PDF report generation tools and event-triggered email notification pipelines, reducing manual administrative effort and demonstrating the ability to translate business process rules into working automation code.

End-to-End Predictive AI Application | Applied AI Module

Python · Keras · TensorFlow · Scikit-Learn · FastAPI · Uvicorn · React · Next.js · Tailwind CSS · joblib

- Architected a complete end-to-end AI application decoupling a React/Next.js frontend from a Python ML backend covering the full build, deploy, measure, and iterate cycle, including model training, API serving, and a dynamic user-facing interface.
- **Engineered a leakage-free ML pipeline:** cleaned and transformed raw datasets, dropped post-event variables to prevent data leakage, and engineered contextual features (temporal data, rolling averages, comparative metrics) to produce a model trained strictly on pre-event information.
- **Evaluated and compared multiple model architectures:** Multilayer Perceptrons (MLP) with Dropout regularisation, 1D Convolutional Neural Networks (CNN) with 3D tensor reshaping, and Gradient Boosted Trees (GBT) selecting the best performer and implementing class_weights to handle severe data imbalance.
- Applied K-Means unsupervised clustering to extract actionable business insights from unlabelled data, validating results with Silhouette Scores and Adjusted Rand Index demonstrating both supervised and unsupervised AI in one project.
- Serialised the trained model with joblib and served real-time predictions via a FastAPI/Uvicorn REST API (/predict endpoint) with graceful error handling and CORS configuration then consumed the API from a Next.js/React frontend with async/await data fetching, loading states, and Tailwind CSS data visualisations.

EDUCATION

MSc Computer Science | Nottingham Trent University, UK — *2025 – Present*

- Modules: AI & Machine Learning, Advanced Software Engineering (CI/CD, Jenkins), Cloud Computing, Distributed Systems, Software Architecture.

BSc Computer Science | University of Mumbai, India — *2022 – 2025*

- Graduated with distinction: A+ (CGPA 9.49 / 10) - equivalent to a UK First-Class / strong 2:1.
- Core modules: Data Structures & Algorithms, Object-Oriented Programming, Database Design, Distributed Systems, Software Engineering.

CERTIFICATIONS

- CS50: Introduction to Computer Science - Harvard University
- Web Applications for Everybody - University of Michigan
- HTML, CSS, and JavaScript for Web Developers - Johns Hopkins University

ADDITIONAL INFORMATION

Right to Work: UK Student Visa - eligible to work up to 20 hours/week during term time; full-time during university holidays.

AI Tools: Daily user of Claude (Anthropic), ChatGPT (OpenAI), Google Gemini, GitHub Copilot, and Cursor using AI to accelerate development, debug complex problems, generate and review code, write test cases, and stay ahead of emerging development practices.

Soft Skills: Problem-Solving, Clean Code Discipline, Proactive Learner, Cross-Functional Communication, Teamwork, Attention to Detail, Time Management.

Languages: English (Professional Working Proficiency), Hindi (Native).

REFERENCES

Available on request.